MATH 464. HW #8. Lazizbek.

Ex. 3.17.

$$(1)\quad \begin{cases} \min\ z = 2x_1 + 3x_2 + 3x_3 + x_4 - 2x_5 \\ \text{s.t.}\quad x_1 + 3x_2 + 0x_3 + 4x_4 + x_5 = 2 \\ \quad x_1 + 2x_2 + 0\cdot x_3 - 3x_4 + x_5 = 2 \\ \quad -x_1 - 4x_2 + 3x_3 + 0\cdot x_4 + 0\cdot x_5 = 1. \end{cases}$$

$$x \geq 0, \quad x \in \mathbb{R}^5$$

Auxilary problem for the given s.f. problem (1) with $\left(\frac{1}{3}\right)$. equality constrains is as follows:

$$(2)\quad \begin{cases} \min\ w = 0\cdot x_1 + 0\cdot x_2 + 0\cdot x_3 + 0\cdot x_4 + 0\cdot x_5 + x_6 + x_7 \\ \text{s.t.}\quad x_1 + 3x_2 + 0\cdot x_3 + 4x_4 + x_5 + 0x_6 + 0x_7 = 2 \\ \quad x_1 + 2x_2 + 0\cdot x_3 - 3x_4 + x_5 + 0\cdot x_6 + 0\cdot x_7 = 2 \\ \quad \left(\frac{-1}{3}\right)x_1 - \frac{4}{3}x_2 + 1\cdot x_3 + 0\cdot x_4 + 0\cdot x_5 + 0\cdot x_6 + 0\cdot x_7 = \frac{1}{3} \end{cases}$$

$$x \geq 0, \quad x \in \mathbb{R}^7$$

Matrix s.f. of the auxilary problem (2):

$$(3)\quad \begin{cases} \min\ w = c_w^T \cdot x \\ \text{s.t.}\quad Ax = b \\ \quad x \geq 0 \\ \quad x \in \mathbb{R}^7 \end{cases}$$

$$c_w^T = [0\ 0\ 0\ 0\ 0\ 1\ 1]^T$$

$$A = \begin{bmatrix} 1 & 3 & 0 & 4 & 1 & 1 & 0 \\ 1 & 2 & 0 & -3 & 1 & 0 & 1 \\ \frac{-1}{3} & \frac{-4}{3} & 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$b = \left[2,\ 2,\ \frac{1}{3}\right]^T$$

(4) choose basis $\beta_1 = \{6, 7, 3\} \Rightarrow$

(5) $\begin{cases} x_B = (x_6, x_7, x_3) = (2, 2, \frac{1}{3}) \geq 0 \\ x_N = (x_1, x_2, x_4\ x_5) = (0, 0, 0, 0) \geq 0 \end{cases} \Rightarrow$

(6) $B = [A_6\ A_7\ A_3] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, so $x_B = B^{-1} b = b = \begin{bmatrix} 2 \\ 2 \\ \frac{1}{3} \end{bmatrix}$

(7) $-z = -c_B^T x_B = -[0\ 0\ 3] \cdot \begin{bmatrix} 2 \\ 2 \\ 1/3 \end{bmatrix} = -1$, recall $\begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{matrix}^T$ $c_z = [2, 3, 3, 1, -2, 0, 0]^T$

(8) $-w = -c_{wB}^T x_B = -[1\ 1\ 0] \cdot \begin{bmatrix} 2 \\ 2 \\ 1/3 \end{bmatrix} = -4$

(9) reduced costs for z:

$$\bar{c}_z = c_z - c_{zB}^T \overbrace{B^{-1}}^{=I} A =$$

$$= [2, 3, 3, 1, -2, 0, 0]^T - [0\ 0\ 3] \cdot I A =$$

$$= [2, 3, 3, 1, -2, 0, 0]^T - [0, 0, 3] \cdot \begin{bmatrix} 1 & 3 & 0 & 4 & 1 & 1 & 0 \\ 1 & 2 & 0 & -3 & 1 & 0 & 1 \\ -\frac{1}{3} & -\frac{4}{3} & 1 & 0 & 0 & 0 & 0 \end{bmatrix} =$$

$$= [2, 3, 3, 1, -2, 0, 0]^T - [-1\ -4\ 3\ 0\ 0\ 0\ 0] = [3, 7, 0, 1, -2, 0, 0]$$

(10)

$$\bar{c}_w = c_w - c_{wB}^T \cdot \overbrace{B^{-1}}^{=I} A = \overset{c_w}{[0\ 0\ \overset{3}{0}\ 0\ 0\ \overset{6}{1}\ \overset{7}{1}]^T} - [\overset{6}{1}, \overset{7}{1}, \overset{3}{0}] \cdot I \cdot A$$

$$= [0, 0, 0, 0, 0, 1, 1] - [1, 1, 0] \cdot \begin{bmatrix} 1 & 3 & 0 & 4 & 1 & 1 & 0 \\ 1 & 2 & 0 & -3 & 1 & 0 & 1 \\ -\frac{1}{3} & -\frac{4}{3} & 1 & 0 & 0 & 0 & 0 \end{bmatrix} = [-2\ -5\ 0\ -1\ -2\ 0\ 0]^T$$

reduced cost for w ↑

(11) $\mathcal{U} = B^{-1}A = A$

So, the initial tableau by (4) - ... - (10), (11):

		x_1	x_2	x_3	x_4	x_5	x_6	x_7	
row ① $-z =$	-1	3	7	0	1	-2	0	0	$\bar{c}_z$
② $-w =$	-4	-2	-5	0	-1	-2	0	0	$\bar{c}_w$
(12) ③ $x_6 =$	2	1	3	0	4	1	$\boxed{1}$	0	
④ $x_7 =$	2	1	2	0	-3	1	0	$\boxed{1}$	
⑤ $x_3 =$	$\frac{1}{3}$	$-\frac{1}{3}$	$-\frac{4}{3}$	$\boxed{1}$	0	0	0	0	

Status:

- $\beta_1 = \{6, 7, 3\}$
- $x = (0, 0, \frac{1}{3}, 0, 0, 2, 2)$
- $z = 1$
- $w = 4$
- Non optimal
- Direction: [ ]
- Valid pivot: ⊙

		x_1	x_2	x_3	x_4	x_5	x_6	x_7
$R1 = r1 + 2 \cdot r4$	3	5	11	0	-5	0	0	2
$R2 = r2 + 2 \cdot r4$	0	0	-1	0	-7	0	0	2
$R3 = r3 - r4$ $x_6 =$	0	0	1	0	7	0	$\boxed{1}$	-1
$R4 = r4$ $x_5 =$	2	1	2	0	-3	$\boxed{1}$	0	1
$R5 = r5$ $x_3 =$	$\frac{1}{3}$	$-\frac{1}{3}$	$-\frac{4}{3}$	$\boxed{1}$	0	0	0	0

Status:

- $\beta_2 = \{6, 5, 3\}$
- $x = (0, 0, \frac{1}{3}, 0, 2, 0, 0) \geq 0$
- $z = -3$, $w = 0$
- Non optimal, $\bar{c}_{w4} = -7 < 0$
- Degenerate $\Rightarrow$ <u>Bland's Rule</u>!

- I'm going to use Bland's Rule by changing basis to go out of degeneracy:

		x_1	x_2	x_3	x_4	x_5	x_6	x_7
$R1 = r1 + \frac{5}{7} r3$ $-z =$	3	5	$\frac{82}{7}$	0	0	0	$\frac{5}{7}$	$\frac{9}{7}$
$R2 = r2 + r3$ $-w =$	0	0	0	0	0	0	1	1
$R3 = \frac{1}{7} \cdot r3$ $x_4 =$	0	0	$\frac{1}{7}$	0	$\boxed{1}$	0	$\frac{1}{7}$	$-\frac{1}{7}$
$R4 = r4 + \frac{3}{7} r3$ $x_5 =$	2	1	$\frac{17}{7}$	0	0	$\boxed{1}$	$\frac{3}{7}$	$\frac{4}{7}$
$R5 = r5$ $x_3 =$	$\frac{1}{3}$	$-\frac{1}{3}$	$-\frac{4}{3}$	$\boxed{1}$	0	0	0	0

Status:

- $\beta_3 = \{4, 5, 3\}$ = Initial b.f. Basis
- $x = (0, 0, \frac{1}{3}, 0, 2, 0, 0) \geq 0$
- $z = -3$, $w = 0 \Rightarrow$ can stop
- Optimal $\bar{c}_z \geq 0$, $\bar{c}_w \geq 0$.
- still degenerate

Yes, we're still at a degenerate point $\boxed{x^* = (0, 0, \frac{1}{3}, 0, 2)}$, possibly there's a way to go out of degenerate point by Bland's Rule with zero step siz at ⊙ I pointed out, but I'm leaving here as my answer is matching with the software, too!